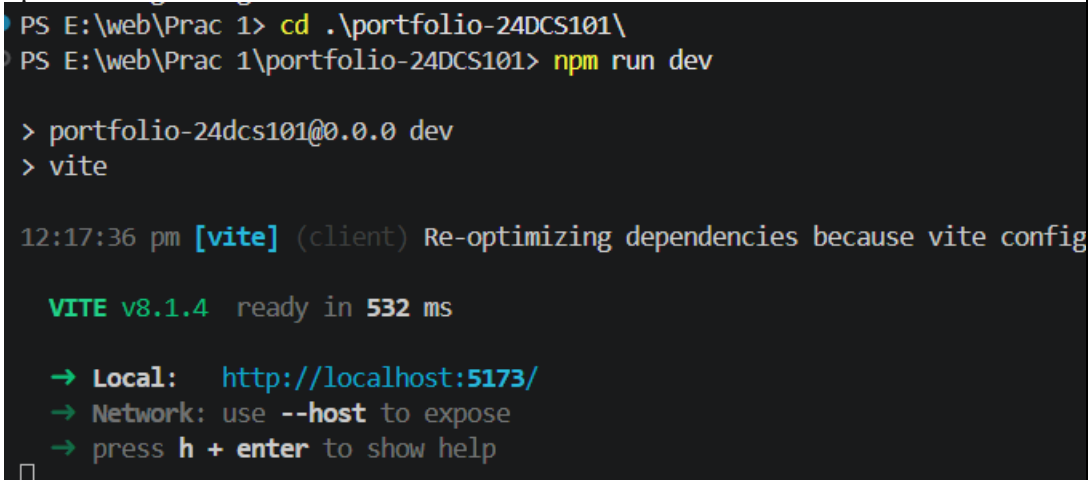


Practical 2 : State Management and Routing in React

Objective	To implement reactive state management using <code>useState</code> and multi-page navigation using React Router in the existing portfolio application.
Problem Statement	• Extend the portfolio application from Practical 1 by adding React Router. • Implement a navigation bar with at least 3 routes: Home, Projects, and Contact. • Each route must render a distinct component without a full page reload. • Add at least 2 <code>useState</code> variables used meaningfully one for toggling UI visibility and one for managing a form input on the Contact page. • The Contact page must include a controlled input that captures and displays user input in real time.
Project Structure	<pre> App.jsx (wrapped in <BrowserRouter>) ├── NavBar.jsx (links to all routes) ├── Route: "/" → Home.jsx ├── Route: "/projects" → Projects.jsx └── Route: "/contact" → Contact.jsx (controlled form, useState) </pre>
Implementation	To Create folder with react and JavaScript write the following command: <pre> npm create vite@latest PORTFOLIO-24DCS101 -- --template react cd PORTFOLIO-24DCS101 npm install npm run dev </pre>  <pre> PS E:\web\Prac 1> cd .\portfolio-24DCS101\ PS E:\web\Prac 1\portfolio-24DCS101> npm run dev > portfolio-24dcs101@0.0.0 dev > vite 12:17:36 pm [vite] (client) Re-optimizing dependencies because vite config changed VITE v8.1.4 ready in 532 ms ➔ Local: http://localhost:5173/ ➔ Network: use --host to expose ➔ press h + enter to show help </pre> Header.jsx: <pre> function Header({ name }) { return (<header className="header"> <h1>{name}</h1> <p>CSE STUDENT </p> </header>); } export default Header; </pre> About.jsx:

```
function About() {
  return (
    <section className="about-section">

      <div className="about-card">

        <h2>About Me</h2>

        <p>
          I'm <strong>Mayur Patel</strong>, a Computer Science student passionate
          about building modern web applications using React, Node.js and
          JavaScript.
        </p>

        <p>
          I enjoy creating clean, responsive user interfaces and continuously
          learning new technologies.
        </p>

      </div>

    </section>
  );
}

export default About;
Skills.jsx:
function Skills() {

  const skills = [
    "React",
    "JavaScript",
    "Node.js",
    "MongoDB",
    "Express",
    "C++",
    "Java",
    "Git",
    "HTML",
    "CSS"
  ];

  return (

    <section className="skills-section">

      <h2>Skills</h2>

      <div className="skills-grid">

        {skills.map((skill) => (

          <div className="skill-card" key={skill}>
```

```
        {skill}

      </div>

    )))}

  </div>

</section>

);

}

export default Skills;

Footer.jsx:
function Footer({ email }) {
  return (
    <footer className="footer">
      <p>{email}</p>
      <p>© 2026 Mayur Patel</p>
    </footer>
  );
}

export default Footer;

Navbar.jsx
import { NavLink } from "react-router-dom";

function Navbar() {
  return (
    <nav className="navbar">

      <div className="nav-links">
        <NavLink to="/">Home</NavLink>

        <NavLink to="/projects">Projects</NavLink>

        <NavLink to="/contact">Contact</NavLink>
      </div>

    </nav>
  );
}

export default Navbar;

App.jsx:
import "./App.css";

import { Routes, Route } from "react-router-dom";
```

```
import Navbar from "../components/Navbar";

import Home from "../pages/Home";
import Projects from "../pages/Projects";
import Contact from "../pages/Contact";
import NotFound from "../pages/NotFound";

function App() {
  return (
    <>
      <Navbar />

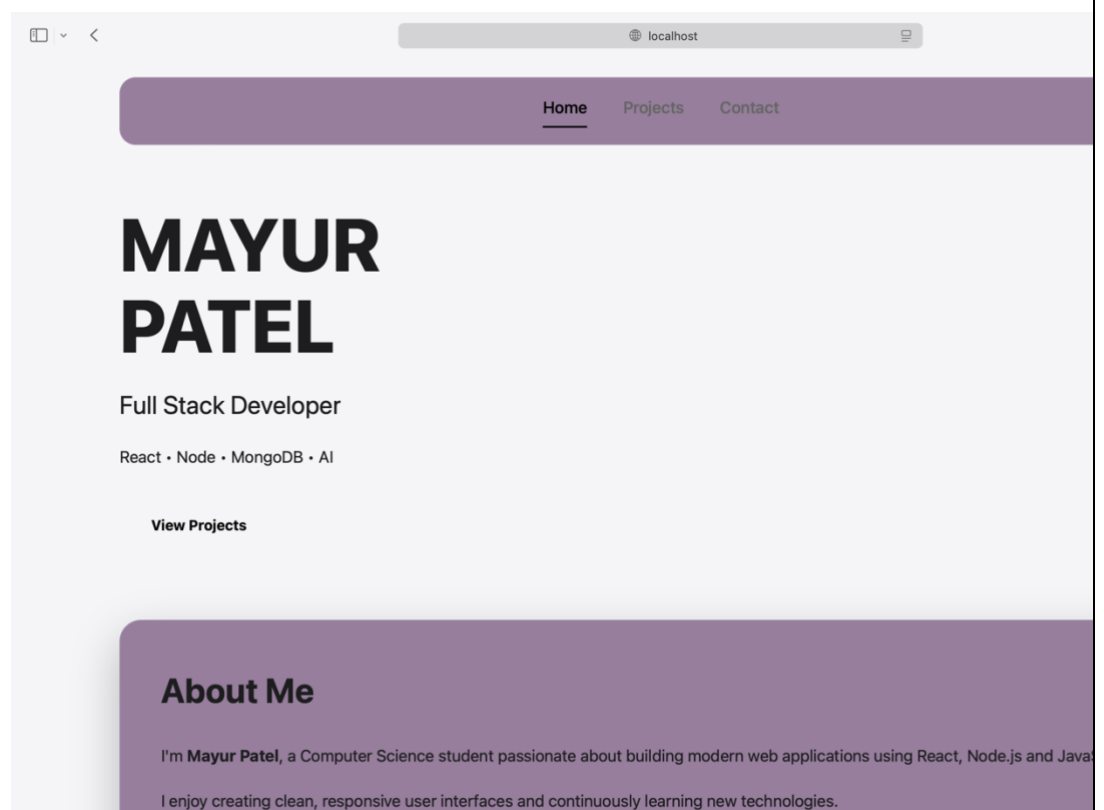
      <Routes>
        <Route path="/" element={<Home />} />

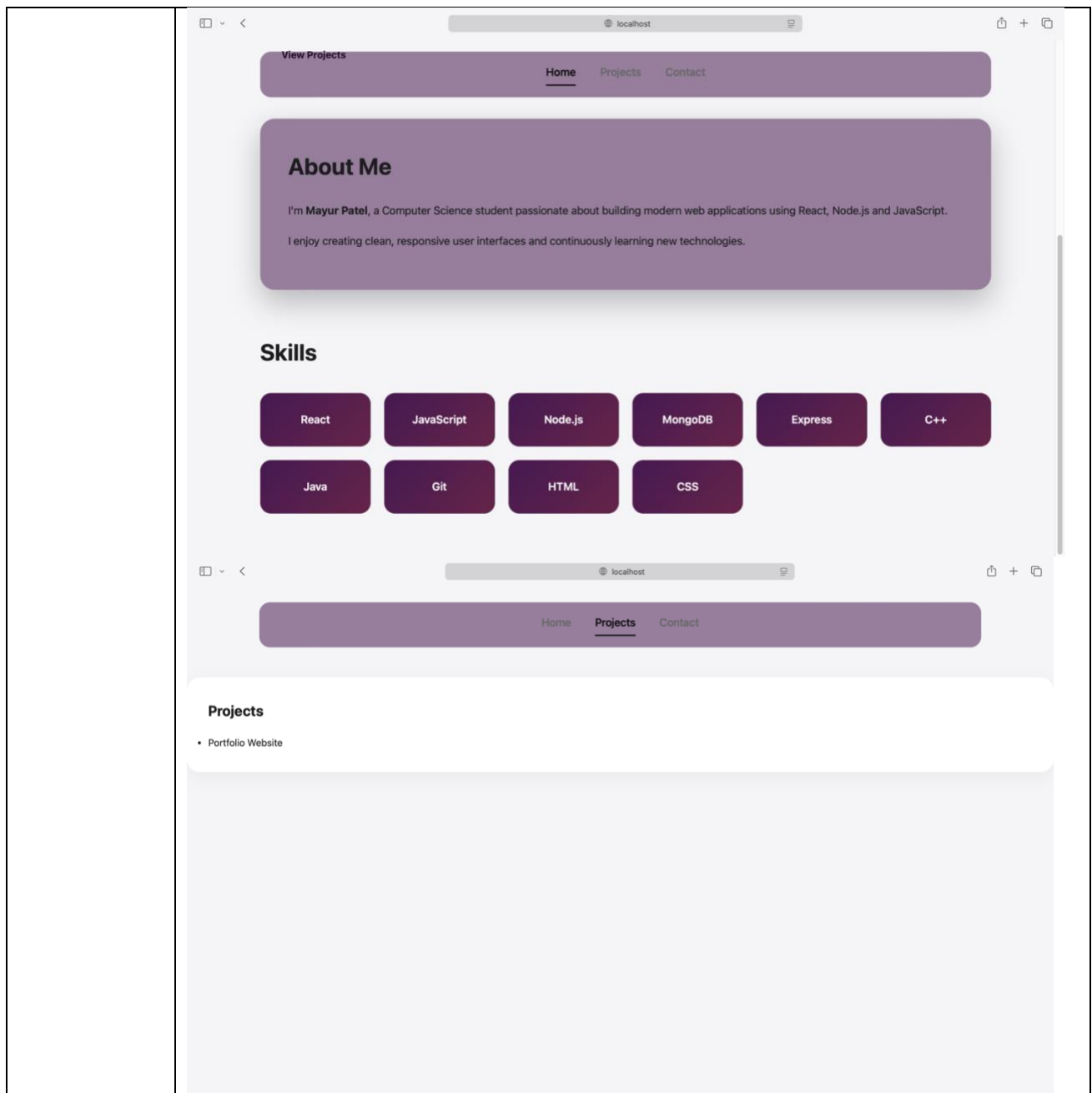
        <Route path="/projects" element={<Projects />} />

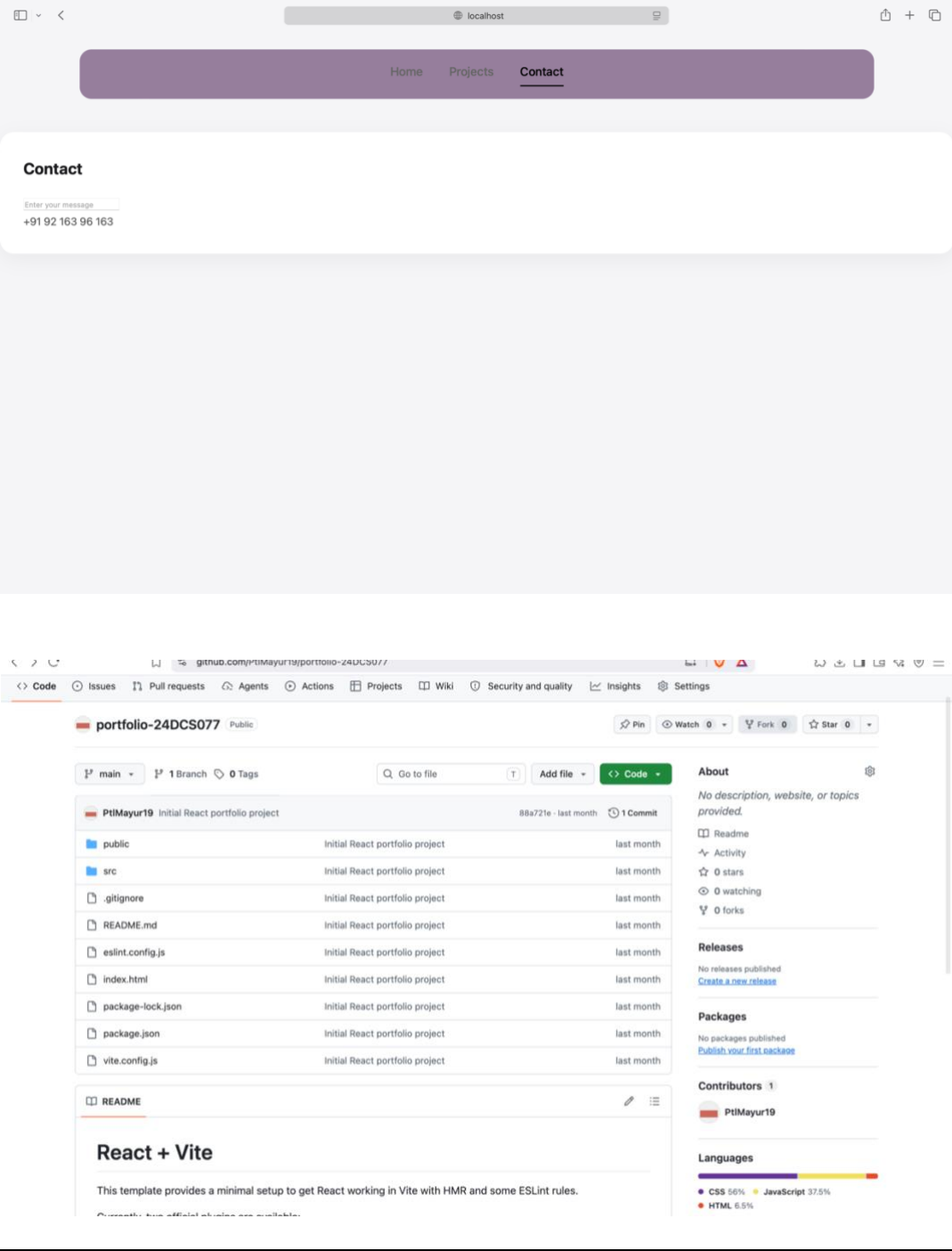
        <Route path="/contact" element={<Contact />} />

        <Route path="*" element={<NotFound />} />
      </Routes>
    </>
  );
}

export default App;
```

Output



	
Conclusion	The portfolio application was successfully enhanced with React Router and useState for dynamic state management. Navigation between Home, Projects, and Contact works without page reloads, while the controlled form and UI toggle demonstrate reactive state handling.